

The Large Angle Pendulum

Dependence of Angular Amplitude on Pendulum Period



Lab 5 Group 9

Aaron Li, Adam Elchaar, Rahul Nanda

The People(s)



Aaron Li (1st Year)

Physics — Experiment Setup, Data
Analysis



Adam Elchaar (1st Year)

Electrical Engineering – Data
Analysis, Presentation



Rahul Nanda (3rd Year)

Computer Science – Background
Physics, Data Collection

Introduction

The purpose of our experiment is to determine the **validity of the small angle approximation** ($\sin \theta \approx \theta$) at various (larger) angles which is used to determine the equation of motion for an undamped pendulum. The small angle approximation simplifies the modeling of a pendulum by enabling the use of simple harmonic motion equations, but we wanted to know when this assumption starts to break down. In other words, when does a pendulum not follow simple harmonic motion (SHM).

Theory: To accomplish this, our experiment measured the variation of period with angular amplitude of the pendulum. As will be explained by our hypothesis, the small angle approximation implies a **constant period** for any pendulum with a constant length, regardless of amplitude.

Method: Using the ESP32 Wrover and ultrasonic sensor (HC-SR04), we collected 5 trials of oscillation data for 5 different amplitudes of the pendulum (25 trials in total).

Data Analysis: Our data analysis attempts to determine the experimental period of the pendulum at 5 different angular amplitudes, and compare it to the expected period according to SHM (small angle approximation) and two methods that do not use small angle approximation to model a pendulum.

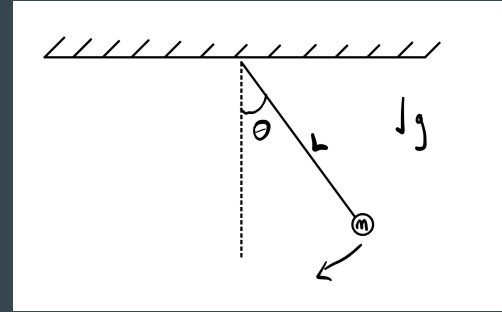
Background Physics – Small Angle Pendulum

A pendulum of mass m under gravitational field strength g , with length L and angular displacement θ can be set up as a system following Newton's second law:

$$\Sigma \vec{F} = m\vec{a}$$

Represented in the system's terms as:

$$-m\frac{g}{L}\sin\theta = m\frac{d^2x}{dt^2} \quad \text{Which can rearrange to} \quad \frac{d^2x}{dt^2} + \frac{g}{L}\sin\theta = 0$$



We show later that for *small angles*, the assumption $\sin\theta \approx \theta$ holds, and hence $\frac{d^2x}{dt^2} + \frac{g}{L}\theta = 0$

Solving this linear differential equation of “Simple Harmonic Oscillation” yields the equation of motion:

$$\theta(t) = A \sin\left(\sqrt{\frac{g}{L}}t + \varphi\right)$$

Which has a period defined by $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{L}{g}}$

Background Physics – Large Angle Pendulum

What if we don't take $\sin(\theta) \approx \theta$ for granted?

Another representation exists: the Taylor Series of expansion where θ centres around 0 approximates this to infinite terms of polynomial:

$$\sin \theta \approx \theta - \frac{1}{3!}\theta^3 + \frac{1}{5!}\theta^5 - \frac{1}{7!}\theta^7 + \dots = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} \theta^{2k+1}$$

Notice that as $\sin \theta$ is small and hence *fractional* in radians, the higher order terms in the polynomial decay away rapidly; leaving $\sin \theta \approx \theta$, as we took for granted previously.

Taking the *elliptic* integral of this helps compute a very familiar, “corrected” form of the period:

$$T = 2\pi \sqrt{\frac{L}{g}} \left(1 + \frac{1}{16}\theta_0^2 + \frac{11}{3072}\theta_0^4 + \dots \right)$$

Background Physics – Numerical Integration

Computational

- 1) Set up the nonlinear differential equation IVP (in python)
 - a) The “step-wise” differential equation solver
- 2) “Step through” using RK-4
 - a) Produces time-series data (like cleaner sensor data)
- 3) Extract Period, Compare results

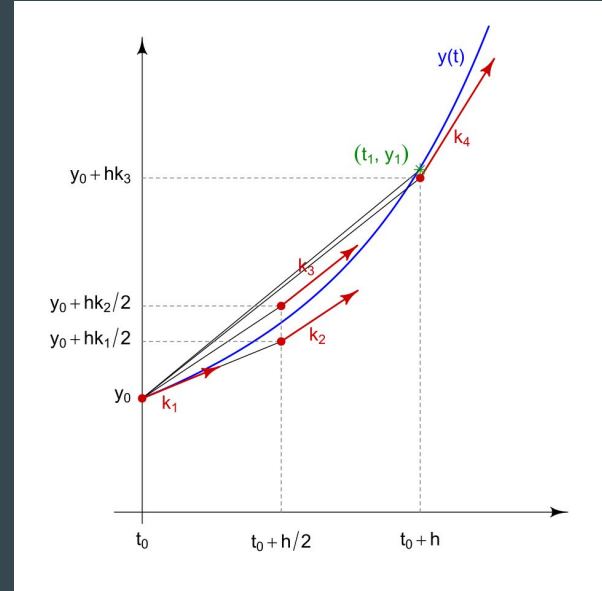


Figure 1: RK4 represented visually

(Wikipedia) “Runge-Kutta Methods”

Experimental Set-Up

5 trials at different release amplitudes

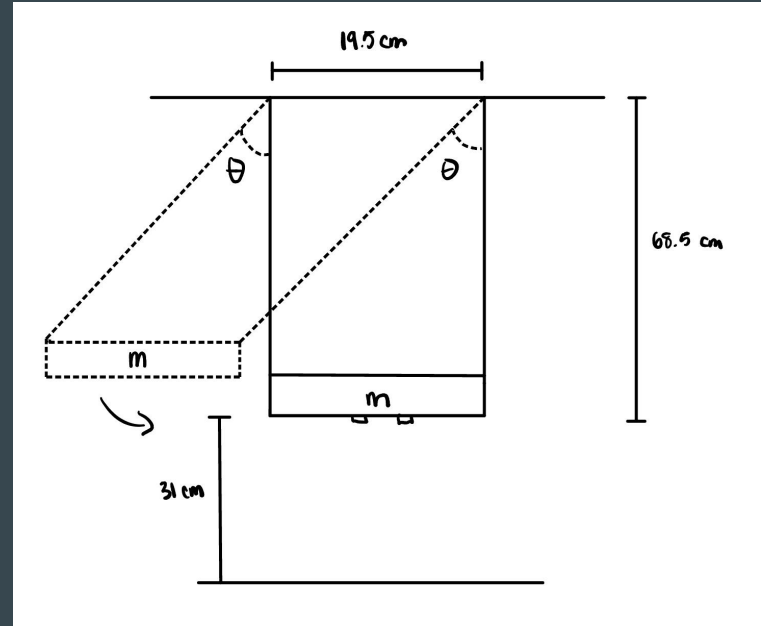
Distance from horizontal rod to ultrasonic sensor (length of pendulum) is **68.5 cm**

Ultrasonic sensor at equilibrium is **31 cm** off the ground

We tested the following 5 release angles (amplitudes): 17.5°, 22.5°, 27.5°, 32.5°, 35.0°

According to the below equation, we calculated what release height these angles would equate to

$$h = 31 + 68.5(1 - \cos \theta)$$



$h = 34 \text{ cm (17.5°), } 36 \text{ cm (22.5°), } 38 \text{ cm (27.5°), } 42 \text{ cm (32.5°), } 44 \text{ cm (35.0°)}$

Oscillation data collected by ultrasonic sensor in arduino

Hypothesis

H_0 : If small angle approximation holds for any set of angular amplitudes, then period will be unchanged across the angular amplitudes.

When $\sin \theta$ can not be approximated to θ , the higher the release angle the greater the period of oscillation will be. Hence, we hypothesize that period will gradually increase with release angle by some (not computed) logarithmic scale, and that three methods of obtaining this period, from (1) experimentation (2) taylor series expansion and (3) numerical integration will produce **mutually coherent** results within 5% error.

Method of Analysis

Materials:

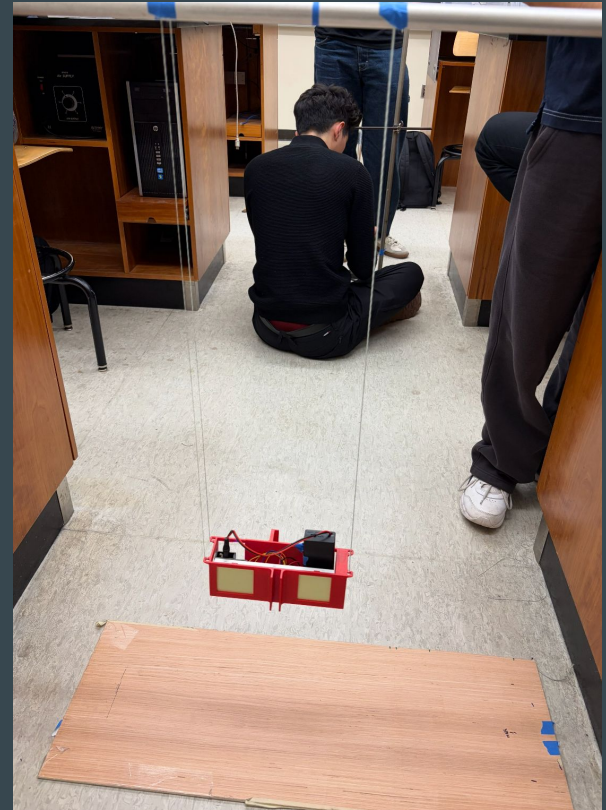
- Arduino ESP32
- Ultrasonic sensor
- 2 long ($> 1\text{m}$) strings
- A rod to which the pendulum can be hung
- Tape

Data Collection:

- At each release angle (amplitude), let ultrasonic sensor collect data for ~ 10 oscillations via window monitor
- Single out a few oscillations for each of the 25 trials (5 for each release angle) which can be analyzed

Data Analysis:

- Python analysis using numpy and matplotlib
- Three methods of determining period from experimental data:
 1. Determining time taken to cross equilibrium point repeatedly
 2. Time taken between minima
 3. Curve-fit the ultrasonic oscillation data, derive period from best-fit equation



Data Analysis – Obtaining Period

Method 1: Between Mean Intersections

```
def line_intersect_method(ria):
    """
    Function to observe np.diffs when the graph crosses the mean point,
    doubled for half-period, quadrupled for full period
    """

    t = ria[:, 0] / 1000.0
    x = ria[:, 1] / 100.0
    mean = np.mean(x)
    y = x - mean
    tol = max(0.1 * np.std(y), 1e-3)
    crosses = []
    for i in range(len(y) - 1):
        if y[i] * y[i + 1] < 0 and (abs(y[i]) > tol or abs(y[i + 1]) > tol):
            frac = abs(y[i]) / (abs(y[i]) + abs(y[i + 1]))
            crosses.append(t[i] + frac * (t[i + 1] - t[i]))
    if len(crosses) < 2:
        return None
    intervals = np.diff(crosses)

    qt_period = np.mean(intervals)
    return float(qt_period * 4), float(qt_period * 2)
```

Method 2: Between Minima

```
def minima_method(ria):
    """
    Function much like line_intersect_method to instead use
    sp.find_peaks on the negative data and observe diffs between
    minima (to take advantage of more stable sensor distances)
    """

    start = 100
    end = 500
    t = (ria[start:end, 0] - ria[start, 0]) / 1000.0
    x = ria[start:end, 1] / 100.0
    ind = sps.find_peaks(-x, distance = 25, height = [-0.32, -0])[0]

    minima_t = []
    for index in ind:
        minima_t.append(t[index])
    if len(minima_t) < 2:
        return None

    intervals = np.diff(minima_t)
    print(intervals)
    half_period = np.mean(intervals)
    return float(half_period * 2), float(half_period)
```

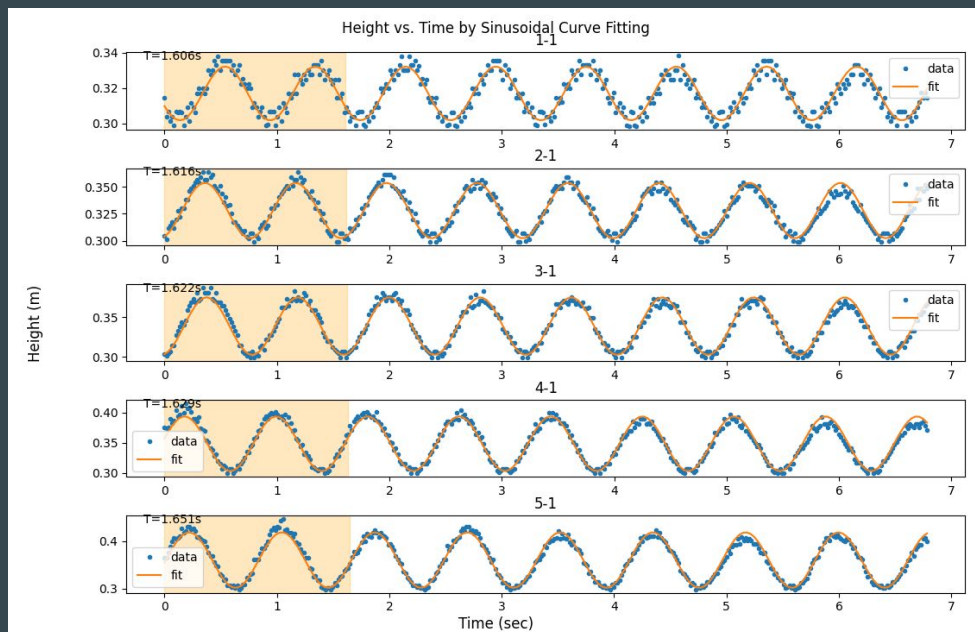
Data Analysis – Winning Method

Method 3: Sinusoidal curve fitting

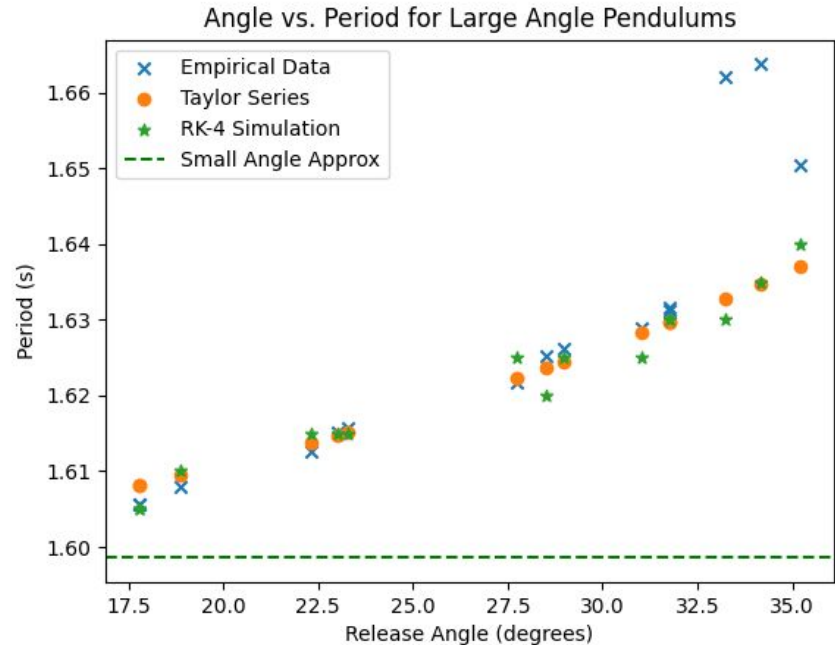
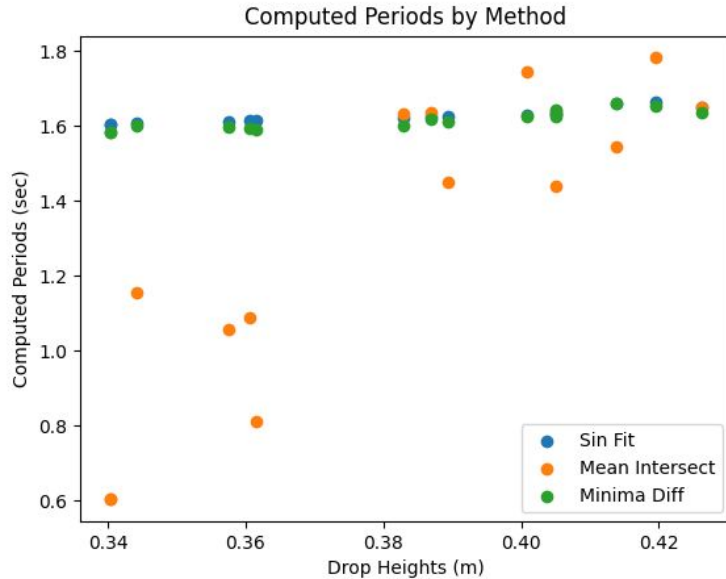
```
start, end = 100, 500
end = min(end, ria.shape[0])
time = (ria[start:end, 0] - ria[start, 0]) / 1000.0
height = ria[start:end, 1] / 100.0

guesses = [amps[idx], 3.78*2, 0, 0.30]

# Create Graph
popt, _ = sp.curve_fit(sin_fit, time, height, p0=guesses)
fit_time = np.linspace(time[0], time[-1], 200)
fit_height = sin_fit(fit_time, *popt)
```



Analysis + Results – Period Comparisons



Conclusion

Through our experiment, we investigated the effect on period depending on release angle as well as the validity of the small angle approximation. Our results showed that for smaller release angles the predicted (through SHM) and measured periods were somewhat similar, but as the release angle increased the measured period started to deviate more and more from the predicted one. This observation is consistent with the small angle approximation because the approximation $\sin\theta \approx \theta$ becomes more inaccurate as θ increases.

As seen in the results, all three methods of computation for the period gave similar results which were all close to the recorded experimental results. This proves our hypothesis since we previously stated they would produce “mutually coherent results”.

In the future, we could take more careful and accurate measurements of the experimental setup and test higher release angles to reduce uncertainty.

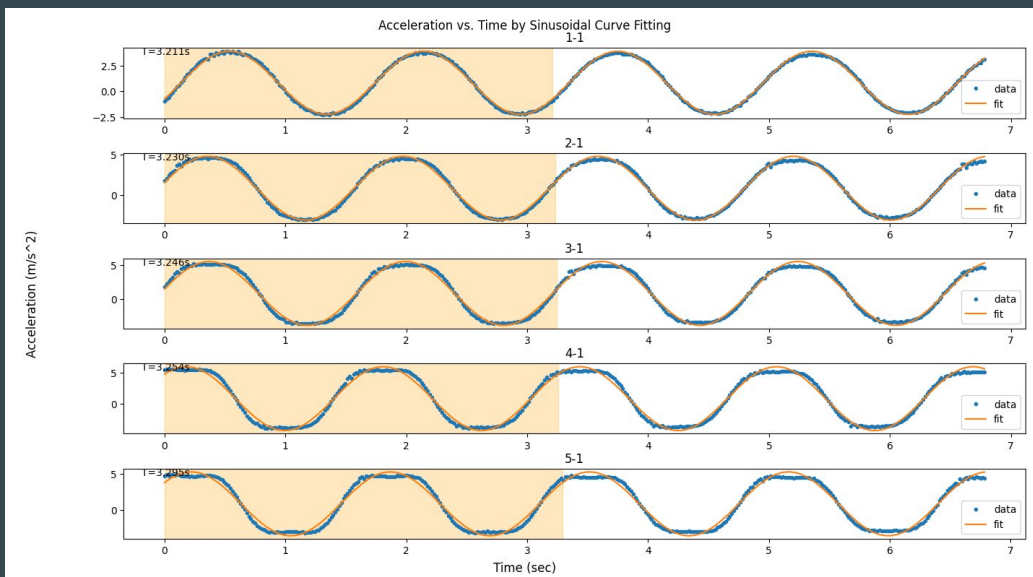
Future Extensions

→ Characterising “hang-time” of the large angle pendulum

- ◆ Seen from our Accelerometer Data (side)

→ “Truly-large” angle pendulums (90-160 degree stiff pendulums)

- ◆ More observable logarithmic scaling



References

Wikipedia contributors. (n.d.). Runge–Kutta methods. In Wikipedia. Retrieved May 30, 2026, from https://en.wikipedia.org/wiki/Runge%E2%80%93Kutta_methods